

# A SIMPLIFIED THEORY OF KINEMATIC WAVES IN HIGHWAY TRAFFIC, PART I: GENERAL THEORY

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**Abstract**—In the theory of “kinematic waves,” as described originally by Lighthill and Whitham in 1955, the evaluation of the shock path is typically rather tedious. Instead of using this theory to evaluate flows or densities, one can use it to evaluate the cumulative flow  $A(x, t)$  past any point  $x$  by time  $t$ . It is shown here how a formal solution for  $A(x, t)$  can be evaluated directly from boundary or initial conditions without evaluation at intermediate times and positions. If there are shocks, however, this solution will be multiple-valued. The correct solution, which is the lower envelope of all such formal solutions, will automatically have discontinuities in slope describing the passage of a shock. To evaluate  $A(x, t)$  at any particular location  $x$ , it is not necessary to follow the actual path of the shock. The solution can be evaluated directly in terms of the boundary data by either graphical or numerical techniques.

## 1. INTRODUCTION

In a famous paper, “On Kinematic Waves,” Lighthill and Whitham (1955) described a theory of one-dimensional wave motion which could be applied to certain types of fluid motion or to highway traffic flow. Richards (1956) independently proposed a similar theory for traffic flow. The key postulate of the (L-W-R) theory was that there exists some functional relation between the flow  $q$  and the density (concentration)  $k$ . The flow is defined for fluids as the rate at which mass passes some point and for traffic as the rate at which vehicles pass some point. The density is defined for fluids as the mass per unit length of channel and for traffic as the number of vehicles per unit length of road. This relation between  $q$  and  $k$  might vary with location  $x$  but not with time  $t$ , i.e.,

$$k(x, t) = k^*(q(x, t), x) \quad (1)$$

or

$$q(x, t) = q^*(k(x, t), x)$$

for some given functions  $k^*$  or  $q^*$ .

The conservation equation (equation of continuity) implies that (with no entering or exiting traffic)

$$\frac{\partial k(x, t)}{\partial t} + \frac{\partial q(x, t)}{\partial x} = 0. \quad (2)$$

which with eqn (1) gives a partial differential equation for  $q(x, t)$ ,

$$w(q(x, t), x) \frac{\partial q(x, t)}{\partial t} + \frac{\partial q(x, t)}{\partial x} = 0. \quad (3)$$

with

$$w(q, x) = \partial k^*(q, x) / \partial q.$$

The  $(1/w)$  is called the “wave velocity,” but in most applications it is more meaningful to describe the time to travel a given distance than the distance traveled in a given time. We

refer to  $w$  as the “pace” of the wave for lack of any other common word for the reciprocal of the velocity.

The usual method for solving eqn (3) is to note that eqn (3) implies that  $q(x, t)$  remains constant along a characteristic curve (wave) for which

$$dt/dx = w(q, x). \quad (4)$$

Thus from any initial or boundary conditions which specify a value of  $q$  at some point  $x_0, t_0$ , the same value of  $q$  applies at all points along the characteristic curve passing through  $x_0, t_0$ , namely the curve

$$t(x) = t_0 + \int_{x_0}^x w(q(x_0, t_0), z) dz. \quad (5)$$

We will be concerned here mostly with the special case of a homogeneous channel or road section for which  $k^*(q, x) = k^*(q)$  is independent of  $x$ . In this case  $w$  is also constant along the characteristic curve and the characteristic curve is a straight line.

One of the complications in the theory results from the possibility that characteristic curves for different values of  $q$  may intersect giving multiples values of  $q$  at the same  $(x, t)$ . To complete the theory, it was necessary to introduce shocks (discontinuities in  $q$ ) and write separate equations for the path of the shock so as to guarantee the conservation of mass across the shock path; the shock itself is not a source of mass. The characteristic curves on either side of the shock path determine the  $q(x, t)$  on each side, but each characteristic terminates at the shock.

It was also necessary to eliminate certain formal solutions of the resulting equations which are inconsistent with a preferred direction of time. The conservation equations are invariant to changing  $x$  to  $-x$  and  $t$  to  $-t$ , but a flow pattern with shocks obtained by reversing the direction of  $x$  and  $t$  is not acceptable.

The numerical solutions of the equations for the shock path are typically rather tedious.

## 2. CUMULATIVE FLOWS

In the analysis of problems in highway traffic flow, particularly “queueing” behind a bottleneck, it is convenient to deal with the “cumulative flow” (see Makagami, *et al.*, 1971). Let

$$A(x, t) = \begin{array}{l} \text{cumulative number of vehicles to pass some location } x \text{ by} \\ \text{time } t \text{ starting from the passage of some reference} \\ \text{vehicle.} \end{array}$$

If vehicles do not pass each other, one could imagine that an observer at location  $x_0$  numbered the vehicles consecutively as they passed him and he attached the numbers to the vehicles. The  $A(x, t)$  then represents the number of the last vehicle to pass an observer at  $x$  before time  $t$ .

The reason that this is so useful for the analysis of traffic flow is that if one draws the curves of  $A(x_1, t)$  and  $A(x_2, t)$  for two locations  $x_1$  and  $x_2$  on the same graph as in Fig. 1, the vertical distance between the curves at time  $t$ ,  $A(x_1, t) - A(x_2, t)$  represents the number of vehicles between  $x_1$  and  $x_2$ , and the horizontal distance between the curves at height  $j$  represents the trip time of the  $j$ th vehicle from  $x_1$  to  $x_2$  if vehicles do not pass each other. The area between the two curves describes the total trip time of all vehicles. Most quantities of interest to the description of traffic flow have simple geometric interpretations.

In Fig. 1, the A-curves should actually be step functions, since the count of vehicles is integer valued. To define a flow, however, one must “smooth” the curve and interpret  $q(x_1, t)$  or  $q(x_2, t)$  as the derivative of the smoothed curve.

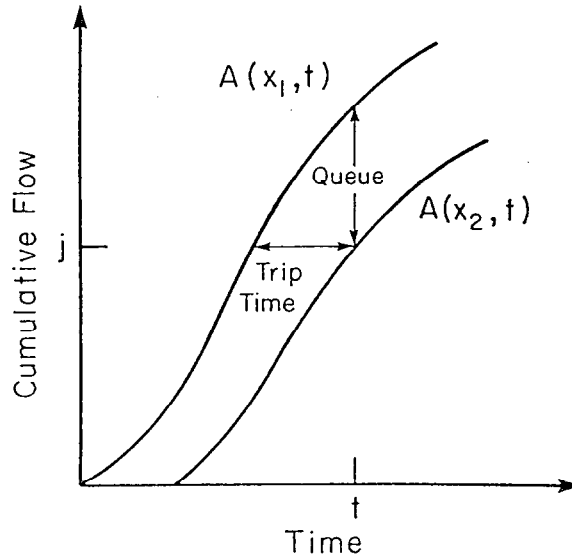


Fig. 1. Cumulative flow at two locations.

One can similarly define  $A(x, t)$  for a fluid as the total mass to have passed  $x$  by time  $t$ , measured relative to the passage of some "streamline," although the physical observables of interest in fluid dynamics are not necessarily analogous to those of interest in traffic flow.

One of the advantages of introducing the function  $A(x, t)$  for either traffic or fluids, is that the existence of such a function itself guarantees the conservation of mass (or vehicle numbers). From the  $A(x, t)$  one can evaluate

$$k(x, t) = \frac{-\partial A(x, t)}{\partial x}, \quad q(x, t) = \frac{\partial A(x, t)}{\partial t}, \quad (6)$$

and the identity

$$\frac{\partial^2 A(x, t)}{\partial x \partial t} = \frac{\partial^2 A(x, t)}{\partial t \partial x}$$

is equivalent to eqn (2) provided that the second derivatives exist. A shock would be represented by a discontinuity in the first derivatives of  $A(x, t)$ , but the conservation equations will still be valid provided only that  $A(x, t)$  is continuous across the shock path, i.e. the number on a vehicle at the shock is the same as viewed from either side of the shock.

The key hypothesis of the L-W-R theory is that eqn (1) is valid at any point where  $q$  and  $k$  are continuous, i.e.

$$\frac{\partial A(x, t)}{\partial t} = q^* \left( \frac{-\partial A(x, t)}{\partial x}, x \right). \quad (7)$$

In the absence of shocks, the formal solution of this equation is obtained in the same way as in the L-W-R theory, namely from the conclusion that  $q$  is constant along the characteristic curves. Knowledge of  $q$  also determines  $k(x, t)$  at all points along the characteristic curve from eqn (1), and together they determine the change in  $A$  along the curve, since

$$dA = (\partial A / \partial x) dx + (\partial A / \partial t) dt = -k dx + q dt = (-k + qw) dx. \quad (8)$$

Thus, if one knows  $A(x_0, t_0)$  and  $q(x_0, t_0)$  at some boundary point, one can determine  $A(x, t)$  at all points along the characteristic curve through  $(x_0, t_0)$ .

In the special case of a homogeneous channel,  $k$ ,  $q$  and  $w$  are all constant along a linear characteristic curve and, as a result of eqn (8), so is  $dA/dx$ . If we interpret  $A(x, t)$  as a surface in a three-dimensional  $(A, x, t)$  space, the surface  $A(x, t)$  is a "ruled surface," a surface generated by a family of straight lines. The value of  $A(x_0, t_0)$  and  $q(x_0, t_0)$  at any boundary point would determine a line in the surface  $A(x, t)$ . From appropriate boundary data one could quite easily construct such a surface and, thereby, determine  $A(x, t)$  everywhere.

### 3. SHOCK CONDITIONS

The main advantage of evaluating the  $A(x, t)$  rather than  $q(x, t)$  relates to the shock conditions. If characteristic curves intersect, the surface  $A(x, t)$  generated in the manner described above is still easily constructed but it gives a multiple-valued function of  $x, t$ . Each of the values of  $A(x, t)$  at the same  $x, t$  is derived from different initial or boundary conditions. To complete the theory, one must specify some recipe which will create from this multiple-valued surface a unique and continuous "solution."

For highway traffic, a driver is influenced mostly by motion of vehicles downstream of him (vehicles of lower number), and the number of vehicles which can pass any point by time  $t$  should not exceed any "constraint" generated by the motion of vehicles downstream at earlier times. If this theory is to make any sense, there will be a natural order of cause and effect relations which will create a "well-posed" problem. We claim that the "solution" one is looking for is the lower envelope of the multiple-valued solution derived from properly set initial or boundary conditions.

The proposed solution is guaranteed to be continuous and therefore satisfy the conservation equations, but the derivatives of  $q$  and  $k$  need not be continuous. A shock path, the path of any discontinuity in  $q$  and  $k$ , will emerge automatically from the geometrical construction of the single-valued  $A(x, t)$ . To determine the time at which a shock passes any location  $x^*$  or the location of a shock at any time  $t^*$ , one need not follow the actual path of the shock. It suffices to construct the graph of  $A(x^*, t)$  versus  $t$  for fixed  $x^*$  directly from the boundary data and see where it has a discontinuity in the slope  $q(x^*, t)$ , or construct a graph of  $A(x, t^*)$  versus  $x$  for fixed  $t^*$  and see where it has a discontinuity in slope  $k(x, t^*)$ .

We have described the solution procedure above as if one were to construct the solution graphically on a continuous space of  $x$  and  $t$ . In doing so, however, one would draw only a certain finite number of characteristic curves and interpolate graphically. If one were to program the procedure for a computer one might prefer to use a discrete lattice of points in the  $x, t$  space. This presents no problem. If one should calculate more than one value of  $A(x, t)$  at some lattice point from different boundary data, one simply chooses the smallest value. It is not even necessary to specify whether or not a shock passed. This can be inferred after one has looked at the resulting solution. In contrast with this, people have encountered all sorts of difficulties trying to solve the original L-W-R equations for  $q$  and/or  $k$  on a lattice.

### 4. BOUNDARY CONDITIONS

There is an enormous variety of practical problems for which the above technique offers a significant simplification as compared with previous methods for solving the L-W-R equations. The present methods were actually devised for the purpose of analyzing flows on a freeway with entrance and exit ramps. One is given a time-dependent rate at which vehicles enter the freeway at entrance  $i$  with destination  $j$  for all  $i$  and  $j$ . The goal is to determine the resulting flow pattern, trip times, exit flows, etc. The details of this will be described in Part III because the present scheme is only a small part of the total problem. The above method will be used to determine  $A(x_{i+1}, t)$  at the  $i + 1$ th ramp from  $A(x_i, t)$  at the  $i$ th ramp or vice versa depending on whether the waves are

traveling forward or backward. This could be done without necessarily also evaluating the  $A(x, t)$  or following the paths of any shocks for intermediate values of  $x$ .

There are a variety of more traditional problems which can be analyzed very easily. For example:

a) One could specify the trajectory of some "lead" vehicle arbitrarily labeled as vehicle number 0 for  $t > t_0$ , or, for a fluid, the trajectory of a piston in a long cylindrical pipe, and also the density  $k(x, t_0)$  for locations upstream of the lead vehicle (or piston) at time  $t_0$ ,  $x < x_0$ .

The trajectory of the lead vehicle is the curve of constant car number  $A(x, t) = 0$ , and the velocity  $v(x, t)$  of the vehicle determines values of  $q$  and  $k$  along the trajectory from eqn (1) and the condition  $q = kv$ . Thus one knows the value of  $A$  and  $q$  along the trajectory from which one can construct a surface  $A(x, t)$  passing through this trajectory.

The initial conditions  $k(x, t_0)$  for  $x < x_0$  determine

$$A(x, t_0) = \int_x^{x_0} k(x', t_0) dx', \text{ with } A(x_0, t_0) = 0$$

and the value of  $q(x, t_0)$ , and therefore a surface  $A(x, t)$  satisfying these initial conditions. The final solution is, of course, the lower envelope of any such surfaces  $A(x, t)$  if there are multiple values.

b) To analyze the flow of vehicles through a traffic signal, one typically specifies the flow  $q(x_0, t)$  at some location  $x_0$  well upstream of the signal, from which one can also evaluate

$$A(x_0, t) = \int_{t_0}^t q(x_0, t') dt'$$

relative to some arbitrary vehicle labeled as vehicle 0 passing  $x_0$  at time  $t_0$ .

At the traffic signal, one further specifies that any vehicle which arrives at the intersection just as the signal turns red will come to a stop and resume its "free speed" soon after the signal turns green. In effect, one specifies the trajectory of a vehicle of known vehicle number, which in turn determines a corresponding surface  $A(x, t)$  as in (a). The final  $A(x, t)$  will be the lower envelope of surfaces determined from the initial flow or from the signal. A shock path defining the back of the "queue" will be the locus of points where these surfaces meet.

c) In the freeway problem mentioned above, one might specify the entrance flow on some ramp at  $x_{i+1}$ , but this flow plus the through flow approaching  $x_{i+1}$  might exceed the capacity of the freeway downstream of the ramp. If the ramp flow does actually pass the "bottleneck," then one knows the maximum through traffic that can pass  $x_{i+1}$ . One will thus know the  $A(x_{i+1}, t)$  for the through traffic from which one can construct a surface  $A(x, t)$  for  $x < x_{i+1}$ . The shock for the back of the queue will be the intersection of this surface with the surface generated from the upstream flows.

## 5. HISTORICAL NOTES

In retrospect, one can't help wonder why something as simple as this has gone unnoticed for more than 35 years or, if it was noticed, that it was not exploited. Almost every book dealing with traffic theory which describes the L-W-R theory not only repeats the same arguments as in the original papers but even reproduces some of the figures from these papers.

The use of cumulative "input-output" curves itself has an interesting history. Edie and Foote (1960, 1961) drew graphs of the cumulative number of vehicles to pass specific locations in the Holland Tunnel. This was perhaps more a convenient way of presenting their experimental observations than the basis of any "theory," although they did use these graphs as a means of showing the propagation of shocks. Herman and Rothery (1965) also used cumulative curves to describe experimental observations and gave geometric interpretations of various derived quantities.

In the "transportation science" literature, the first use of cumulative curves as a tool for making theoretical prediction seems to have been by Gazis and Potts (1965). These curves were described and used with no references, as if they were an obvious tool. In the "traffic engineering" literature, cumulative curves are described in a paper by Moskowitz and Newman (1963), but it was apparently a "standard procedure" for the analysis of freeway traffic among the traffic engineers at the California Division of Highways (now Caltrans). Some of the engineers who were with Caltrans at that time say that "everyone knew it." It seems likely that this was an invention of the late Karl Moskowitz.

For the present generation of transportation engineering students, the use of such curves is well-known, but it took a long time (a "generation") for this to become one of the standard tools of traffic engineering. For people working in related fields of queueing, inventory, production, etc., which also deal with the flow of objects past various restrictions, there is still considerable reluctance to exploit simple graphical methods of analysis based on cumulative flows.

The three-dimensional representation  $A(x, t)$  was described by Makigami *et al.* (1971). I first wrote some notes about this for a course on queueing theory to show that an observer at any location  $x$  could construct a curve of cumulative flow, and that the collection of all such curves would describe a surface  $A(x, t)$ . Y. Makigami, a student in this class, decided to do an individual study to explore it further. He even made a plaster of Paris model of a hypothetical traffic pattern.

After we had circulated a draft of a paper, Richard Rothery pointed out that he had described a similar three-dimensional construction in his (unpublished) Ph.D. thesis (Rothery, 1968). We agreed to submit the paper with three authors and make reference to Rothery's thesis.

Rothery's thesis also points out that a stationary flow pattern would give a planar surface for  $A(x, t)$  and that a "transition" from one constant flow pattern to another would propagate with a certain speed. There is no reference in Rothery's thesis, however, to the L-W-R theory or to "waves" or "shocks." Most of Rothery's thesis deals with car-following theories.

At that time Karl Moskowitz also called me on the telephone to announce that he had, for some time, used such three-dimensional figures to describe freeway traffic patterns. I asked him if he could give me a reference, but he said that he never published it or even wrote any technical reports about it. I had no doubt that he had done it, and I should have suggested that he join as another coauthor. I lost an opportunity then to give some recognition to one of the great pioneers of traffic engineering. Unfortunately, he published very little of what he knew.

The three-dimensional representation  $A(x, t)$  was just one of several illustrations of "tandem queues" included in the second edition of my book on queueing theory, (Newell, 1982). Other illustrations involved "blocking" due to a finite storage between "servers," for which the cumulative departures at any server might be constrained by conditions either upstream or downstream. The actual cumulative curve was the smallest of those determined by an constraint.

There was also a simple model of a freeway diamond interchange for which the highway section between an off-ramp and an on-ramp could store only a specified number of cars. Again, the cumulative flow at the off-ramp was constructed as the minimum of an arrival curve determined from upstream conditions and a departure curve determined from downstream conditions at a bottleneck. It was not explicitly noted that when these two curves crossed, it was the time when the shock wave passed; nor was there any mention of the connection with the L-W-R theory, but it was noted that one did not need to know the details of the traffic flow pattern at intermediate points.

More recently I derived some "backward" solutions of the L-W-R equations in terms of the  $A(x, t)$  but with no shocks (Newell, 1988). I am embarrassed that I did not recognize then the simple solution for the shock conditions. The emphasis in this paper was on other things (which were less interesting).

The equations of "kinematic waves" arise in many other applications, particularly in fluid dynamics (flood waves, bores, sonic booms, etc.). I made some inquiries among

fluid dynamics specialists if analogous methods of determining shock paths were well-known in fluid dynamics, and then sent some notes on this to G. B. Whitham. He replied that the shock-fitting is "well-known to us," which he later explained meant "to me, my students, and other hangers-on."

Whitham sent me a reprint of a paper by J. C. Luke (1972) on erosion of soil. In this paper the analogue of the  $A(x, t)$  is the elevation of the ground. Luke starts with equations analogous to eqn (7) and solves these equations by, in effect, going back to the kinematic wave equations (3) to apply the method of characteristics. The shock conditions, however, are determined from the lower envelope of  $A(x, t)$  surfaces as in Section 3.

It would seem that this trick is obvious to some people working in continuum mechanics but unknown to others, much as the use of cumulative curves for the analysis of traffic problems was obvious to some people and unknown to others.

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